2. Find x such that $\cos(x) + \sin(x) = 0$.

A. $x = \frac{\pi}{4}$

B. $x = 0$

C. $x = \pi$

D. $x = \frac{\pi}{2}$

E. $x = \frac{3\pi}{4}$

3. Consider the equation $\sin(x) = 2 - k$, with $0 \leq x \leq \pi$. This equation has *at least* one solution if and only if:

A. $k \geq 1$

B. $1 \leq k \leq 2$

C. $k \leq 2$

D. $-1 \leq k \leq 1$

E. $1 \leq k \leq 3$